Supplementary Material for Mitigating the Rationale-Prediction Conflict for Data-Centric Rationalization

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***Abstract*—This supplementary material provides content for reproducibility verification and supporting arguments that underpin the claims made in the main paper, but it is not essential for the primary exposition. Building upon the research framework introduced in the main paper, BINGO, this supplement elaborates on the following: Run and Artifact Provenance (Sect. I), Proof Sketch and Details (Sect. II), the definition of gradient conflicts (Sect. III), additional empirical results (Sect. IV), comparisons of algorithmic backbones (Sect. V), further experimental details (Sect. VI), and the potential impact of rationalization for data management in the era of large language models (Sect. VII).**

I. RUN AND ARTIFACT PROVENANCE

We release the artifact to facilitate review and reproduction. The code and data are available at: **GitHub Repository**. The supplementary material is also included in the repository as `supplemental_material.pdf` for review purposes. Reviewers are kindly referred to the `README.md` file for detailed instructions on how to reproduce the results.

II. PROOF SKETCH AND DETAILS

In this section, we provide a detailed characterization of the disjoint optimal solutions for the prediction and rationale objectives in rationalization.

Let $M \in \mathcal{M}$ denote the decision variable, where $\mathcal{M}$ is the feasible solution space. Let $F_a(M)$ represent the prediction objective and $F_b(M)$ represent the rationale objective. Denote their corresponding sets of globally optimal solutions by

$$M_a^* = \arg\min_{M\in\mathcal{M}} F_a(M), \quad M_b^* = \arg\min_{M\in\mathcal{M}} F_b(M). \tag{1}$$

Proposition 1: If the optimal solutions of the prediction and rationale objectives satisfy $M_a^* \neq M_b^*$ and $M_a^* \cap M_b^* = \varnothing$, then the corresponding learning objectives F_a and F_b are said to be conflicting.

PROOF. Assume, for contradiction, that F_a and F_b are not conflicting. Then there exists at least one feasible solution $M_0 \in \mathcal{M}$ that simultaneously achieves the global optimum of both objectives. By the definition of optimal solution sets, we have $M_0 \in M_a^*, M_0 \in M_b^*$. Therefore, $M_0 \in M_a^* \cap M_b^*$, which implies $M_a^* \cap M_b^* \neq \varnothing$. However, according to the premise, $M_a^* \cap M_b^* = \varnothing$, which leads to a contradiction. Hence, the assumption that F_a and F_b are not conflicting is false. Therefore, objectives F_a and F_b are conflicting.

In the above analysis, we have the premise that $M_a^* \cap M_b^* = \varnothing$. According to the analysis in Sect. III-C, the optimal solution of the prediction objective in rationalization tends to select the entire input text X. Therefore, we have $M_a^* = \{M_1\}$, where $M_1 = \{m_1, m_2, \dots, m_l\}, m_i \in \{1\}(1 \leq i \leq l))$, i.e., a length-$l$ all-one selection vector. In contrast, the rationale objective encourages selecting a subset of tokens with size $s \cdot l$, where $s < 100\%$. Under the coherence constraint, the optimal rationale set is given by $M_b^* = \{M_2, M_3, \dots, M_n\}$ $(n > 1)$, where each $M_i \in \{0,1\}^l$ satisfies $\sum_j M_i^{(j)} = s \cdot l$. Since $s < 100\%$, it follows that $M_1 \notin M_b^*$. Consequently, we obtain

$$M_a^* \cap M_b^* = \varnothing. \tag{2}$$

Algorithm 1 Adam [2]

Require: parameters θ_0;
Require: Stochastic objective function with parameters $f(\theta)$;
Require: learning rate α;
Require: decay rates $\beta_1, \beta_2 \in [0, 1)$;
Require: small constant ϵ.
Initialize first moment and second moment: $m_0 \leftarrow 0$, $v_0 \leftarrow 0$
for $t = 1$ to T **do**
  Compute stochastic gradient: $g_t \leftarrow \nabla_\theta f(\theta_{t-1})$
  $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$
  $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$
  $\hat{m}_t \leftarrow m_t / (1 - \beta_1^t)$
  $\hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$
  $\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$
end for
Return: θ_T (Resulting parameters)

III. DEFINITION OF GRADIENT CONFLICTS

Inspired by Yu et al. [1], we design a conflict detection method based on gradient consistency to quantify the alignment between rationale and prediction objective during training. We define the definitions as follows:

Definition 1: We define θ_{ij} as the angle between two gradients $\boldsymbol{g}_i$ and $\boldsymbol{g}_j$. We define the gradients as conflicting when $\cos\theta_{ij} < 0$. In this paper, we denote the angle between prediction and rationale gradients by θ.

Definition 2: We define the gradient magnitude similarity between two gradients $\mathbf{g}_i$ and $\mathbf{g}_j$ as $\Phi(\boldsymbol{g}_i, \boldsymbol{g}_j) = \frac{2\|\boldsymbol{g}_i\|_2 \|\boldsymbol{g}_j\|_2}{\|\boldsymbol{g}_i\|_2^2 + \|\boldsymbol{g}_j\|_2^2}$.

IV. Additional Empirical Results

As shown in Sect. III-C in the main paper, we empirically validate the existence of rationale–prediction conflicts during the dynamic training process. In addition to the Appearance dataset, we also conduct experiments on the Aroma dataset and the harder Palate dataset. As shown in Fig. 1, and Fig. 2, we provide the gradient conflict visualization between rationale gradients and prediction gradients. For the rationale gradients, we focus on the gradients of coherence and sparsity constraints that are widely studied in rationalization tasks [3]–[6].

V. Comparisons of Algorithmic Backbones

To better illustrate the novelty of our proposed method, we provide an intuitive overview of the widely used Adam optimization process. As shown in Algorithm 1, Adam iteratively computes stochastic gradients, updates the first-order moment (capturing the mean of past gradients) and the second-order moment (capturing the variance), applies bias correction, and finally updates parameters with adaptive step sizes. The detailed algorithm of Bingo, which extends this process to explicitly address rationale–prediction conflicts, can be found in Sect. IV-C of the main paper.

VI. Further Experimental Setups

Details of Skewed-Explainer. The skewed-explainer synthetic experiment was first proposed by [5], which is designed to deliberately induce degradation in the explainer, to evaluate whether the explainer in the RNP framework can still identify rationales under skewed conditions. Following previous research [7], [8], we first pretrain the explainer using the text classification labels as a masked label for the first token separately. In this way, the predictor will be able to know the category of a rationale merely through whether the first token is selected as part of the rationale or not and may overfit to this positional bias. Therefore before we cooperatively train them, the explainer is initialized with the intentionally misleading pretrained parameters.

We compare the performance of Adam and our proposed Bingo on the Palate dataset as [5] did, as Palate is relatively difficult to learn compared to other datasets [9]. Following previous work [8], we also adopt the notation "Skewk", where k denotes the threshold of the skew: the explainer is pretrained for a few epochs as a special classifier on the first token until its prediction accuracy exceeds k. A higher "Skewk" indicates that the explainer is easier to degenerate.

VII. The Potential Impact of Rationalization for Data Management in the Era of Large Language Models

In contrast to model-centric explanation methods, data-centric approaches focus on identifying patterns inherent in the data itself. From this perspective, rationalization serves not only as an explainability tool but also as a mechanism for data cleaning [11]. This is particularly important for domain-specific large language models (LLMs), whose fine-tuning datasets often contain undesirable artifacts such as biases and

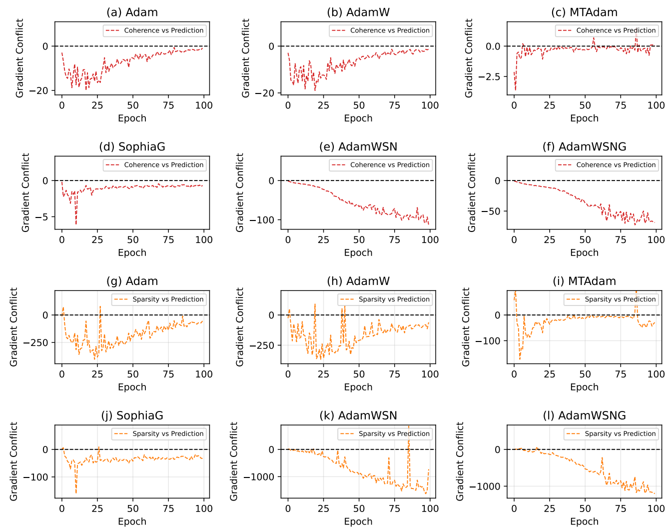

Fig. 1. The gradient conflict on the Beer-Aroma dataset. Following previous research [1], [10], we use the same method to detect gradient conflicts. The red curves illustrate conflicts between the coherence gradient and the target-label gradient during training, while the orange curves indicate conflicts between the sparsity gradient and the target-label gradient.

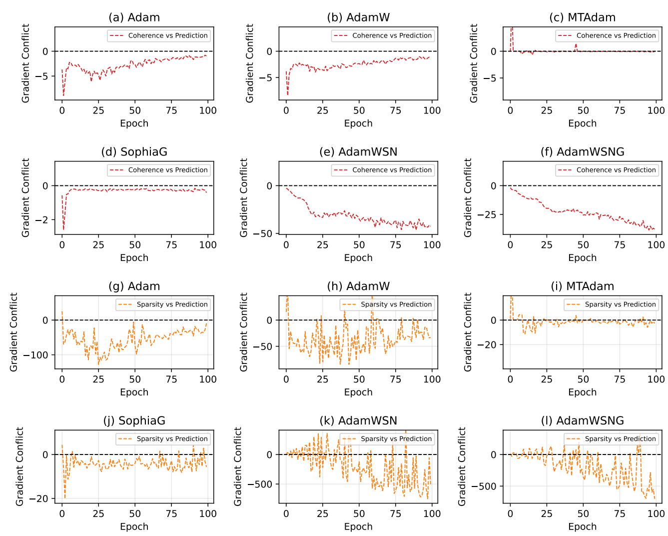

Fig. 2. The gradient conflict on the Beer-Palate dataset. Following previous research [1], [10], we use the same method to detect gradient conflicts. The red curves illustrate conflicts between the coherence gradient and the target-label gradient during training, while the orange curves indicate conflicts between the sparsity gradient and the target-label gradient.

stereotypes [12]. Previous work shows that rationale-based predictors can suppress irrelevant or harmful information, leading to improved robustness and generalization [13]–[15].

Given that small models are often sufficient for rationale extraction and are more efficient for dataset-specific training, a practical pipeline is to first train a small model to identify rationales and then use the resulting rationalized data to fine-tune larger models. Such a strategy can reduce the risk of transferring harmful patterns from noisy training data while simultaneously shortening inputs and lowering memory requirements during fine-tuning [16]. Recent studies suggest that even small models trained solely for data selection, without explicit rationale extraction, can identify high-quality subsets that effectively support LLM fine-tuning [17], [18].

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